

HCF Gas Line Absorption Mitigation: On the Interplay between Pre-emphasis and SOA Nonlinear Compensation

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Abstract We experimentally and numerically investigate the interplay between GLA-dip emphasis (GE) and SOA-booster nonlinear compensation in 140 GBaud PCS-64QAM transmission system. Results show that the interplay causes 0.3 dB SNR penalty and the partial-GE is more favorable, providing 0.5 dB SNR gain at 7-dB dip. ©2026 The Author(s)

Introduction

Hollow-core fibers (HCF) have become one of the most promising technologies to cope with the increasing traffic demand (i.e., in DCI/DCN) thanks to their low attenuation, reduced latency, negligible nonlinearity, etc. [1–3]. Due to their hollow nature, gases such as H_2O , CO , CO_2 can infiltrate into their cores. These gases cause frequency-selective fading, an effect typically named as gas line absorption (GLA)-induced fading (a.k.a. GLA dip), resulting in degradation of SNR and channel throughput [4, 5]. Hence, it is imperative to mitigate GLA-inherent penalties in order to maximize HCF benefits. Gas purification and pressurization have been demonstrated [6] at the cost of a long procedure for >5-km fiber lengths. Digital signal processing (DSP) and/or waveform designs can be used to compensate for GLA, leveraging the advent of coherent technology [7–11]. In particular, entropy-loaded OFDM to mitigate GLA has been studied in [7]. Refs [8–10] proposed adjusting digital subcarrier multiplexing (DSCM) signal frequencies and adding guard bands at GLA positions, at the cost of more transmitter bandwidth requirement. Moreover, GLA emphasis at the transmitter (Tx) side has also been studied in [9], improving the transmission distance. In [11], GLA emphasis at the receiver (Rx) side is numerically investigated to aid the equalizer in reducing the required number of taps.

In this paper, we experimentally investigate the use of GLA emphasis (GE) to mitigate GLA penal-

ties in 140-GBaud PCS-64QAM single-carrier transmission systems over 20 km HCF. Unlike [11], which was limited to numerical study of Rx-side GE only or [9] with Tx-side GE investigation, we experimentally investigate both Tx and Rx GE. We further study the benefit of GE when the Tx is delivering high output power, which requires the integrated SOA booster in HB-CDM to operate in the nonlinear (NL) regime [12, 13] and the use of NL compensation to mitigate the SNR degradation due to SOA nonlinearities. It reveals that the interplay between GE and SOA NL despite including its compensation shows optimal GLA mitigation gain at partial GE as well as larger gap to ideal compensation. Such interplay is confirmed by performing numerical simulation, which allows to extend our study for larger dip levels.

Experimental Setup

We investigate GLA mitigation techniques for the transmission link using the setup shown in Fig. 1(a). A 140 GBaud PCS-64QAM with an entropy of 5.7 (corresponding to 1.2T net capacity, assuming 25% FEC overhead and 3% pilot/protocol overheads) is generated and passed through different Tx DSP blocks (Fig. 1(c)). These includes GLA emphasis (if enabled), Tx S21 pre-emphasis and SOA pre-compensation (if operating in high-power mode). After that, the digital signals are resampled to the DAC rate before being loaded into it. Note that GE is implemented in the frequency domain to facilitate integration with other existing blocks. To focus on the benefits of using different

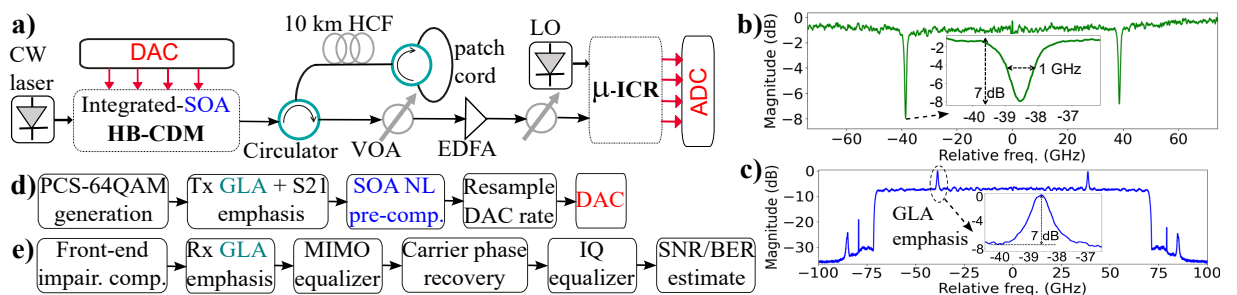


Fig. 1: (a) Experimental setup of 140 GBaud PCS-64QAM signals transmitted over 20-km HCF with realistic transceiver (HB-CDM and μ -ICR); (b) Measured GLA spectrum dip after 20-km HCF; (c) Tx spectrum with 7-dB GE; (d) Tx and (e) Rx DSPs.

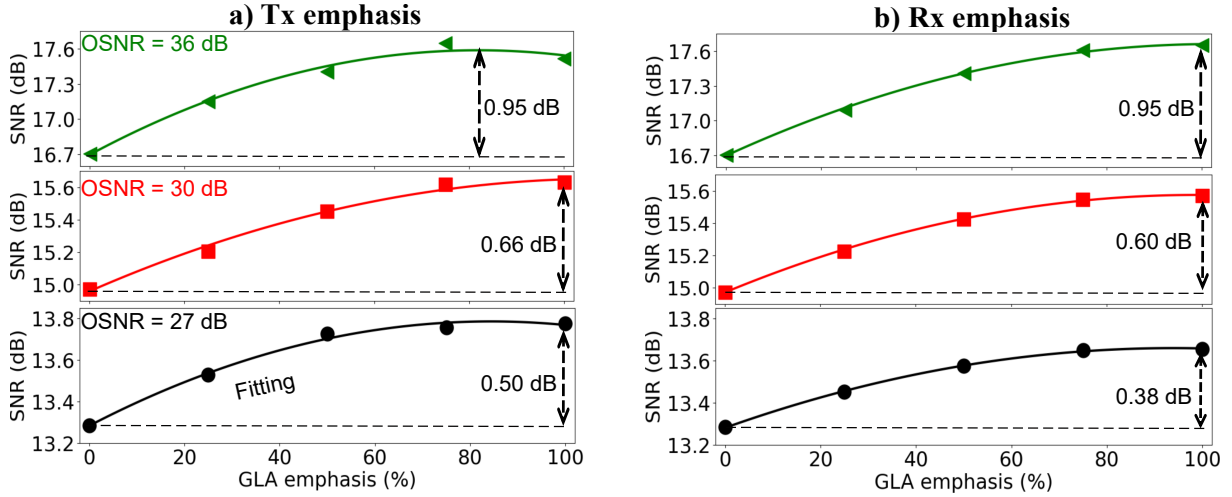


Fig. 2: Measured GLA mitigation at different OSNR (in 0.1 nm) and 7 dB dip, using emphasis at (a) Tx and (b) Rx side.

GEs, the FFT size is set to 2048, which corresponds to a resolution about 100 MHz. The DAC electrical signals are connected to an HB-CDM in order to modulate a laser at 1565 nm, which is the wavelength with stronger dips because of CO . The HB-CDM includes an integrated SOA, whose gain can be adjusted to operate at different output power modes. Since the HCF spool is 10 km, we use circulators to double the transmission distance and to obtain deeper dips. An example of the dip spectrum is shown in Fig. 1(b), with 2 dips of 7 dB depth and 1 GHz full width at half maximum after 20 km HCF. The Tx spectrum with GE is plotted in Fig. 1(c), showing two narrow pre-emphasis spectral peaks. The optical signal power after transmission is adjusted by a variable optical attenuator (VOA), followed by an EDFA in order to vary the OSNR (computed in 0.1 nm). Another VOA is placed after the amplifier to fix the received power into the μ -ICR. The electrical signal outputs of the μ -ICR are quantized by the ADCs for further DSP.

The Rx DSP (Fig. 1(d)) includes Rx front-end impairment compensation, Rx GE (if enabled, with similar FFT resolution as the Tx side), MIMO equalization, carrier phase recovery (carrier frequency offset and laser phase noise compensation), and IQ equalization. Finally, the SNR performance is estimated to compare different GLA mitigation strategies. Note that no CD compensation is used, since the CD of HCF is small and is handled by MIMO equalization. The GLA dip can also be compensated by increasing the number of equalizer taps at the cost of higher complexity [11, 14]. To be realistic for commercial transceivers, in which the number of equalizer taps is limited, we set the memory of equalizer to 51 taps.

In the following, we investigate GE under low-power and high-power SOA modes, which provide -3.1/6.5 dBm Tx output powers, respectively. Note that high Tx power mode is important for applica-

tions such as amplifierless scenarios [13] or high Tx OSNR requirements [2]. To boost Tx power, the integrated SOA operates in the NL regime, and its NL behavior can be compensated based on SOA parameter estimation [15, 16]. The GE performance is compared to back-to-back (B2B) performance, which is measured with a similar setup by replacing the 10 km HCF spool and circulators with an SSMF patch cord.

GLA Emphasis at Tx and Rx Sides

In the presence of a 7 dB GLA dip (after 20 km), we investigate the GE technique at low-power mode in Fig. 2. The SNR versus different GE levels at three OSNRs is plotted when GE is applied at the Tx (Fig. 2(a)) and at the Rx (Fig. 2(b)). Note that the GE profile is estimated in advance using either a high-resolution OSA or by comparing received signal spectra between without and with GLA dips; hence, only amplitude GE is obtained. The phase is then generated from the obtained amplitude information using the Hilbert transform [11].

It is observed that near 100% GE (equivalent to 7 dB) provides the maximum gain regardless of OSNR. The gain decreases with lower OSNR due to more dominant ASE noise. At 27 dB OSNR, Tx GE provides about 0.1 dB more gain than Rx GE, as Rx GE is less effective due to Rx noise enhancement. Hence, for long transmission (equivalent to low OSNR), Tx GE could be preferable.

SOA NL Compensation and GE Interplay

We investigate the Tx GE benefit at low-power and high-power modes, corresponding to without and with SOA NL, at 36 dB OSNR in Fig. 3. In low-power mode, the GE is efficient and the penalty is <0.1 dB compared to B2B. In high-power mode, the optimal SNR gain is obtained when GE ratio is shifted back to around 60% (corresponding to about 4 dB GE). Up to 0.3 dB SNR penalty compared to B2B is observed, indicating the interplay between SOA NL and GE. Finally, we present the SNR as the function of OSNR without GE and

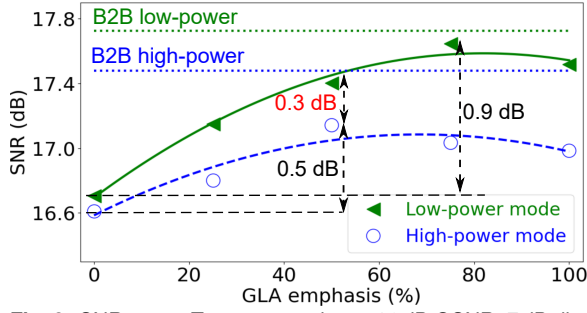


Fig. 3: SNR at two Tx-power modes at 36 dB OSNR, 7 dB dip.

with GE in Figs. 4(a) (low-power mode) and 4(b) (high-power mode). In low-power mode, there exist <0.1 dB SNR penalties compared to B2B performance. In high-power mode, the SNR penalty becomes bigger (i.e., 0.3 dB SNR penalty at 36 dB OSNR) due to the interplay. Due to the relative large amount of power allocated for narrow-band parts of the signal spectrum, the SOA nonlinearities are enhanced. Since SOA nonlinear compensation is not ideal, large GE are more affected when operating in high-power mode.

Numerical Simulations

Since the interplay may be limited by transceiver noise and/or dip level, we carry out the simulations with variable dip levels using the simulation setup, shown in Fig. 5(a). The AWGN noises are added at Tx and Rx sides to emulate the transceiver noise and match the experimental noise level. A laser phase noise of 50 kHz total linewidth and a carrier frequency offset of 0.3 GHz are added in simulations. The SOA is emulated based on parameters estimated from experiment [15]. The GLA emulation is based on the model in [11] with dip characteristics matching the experimental measurement (Fig. 1(b)). The DSP is similar as used in experiment. Fig. 5(b) presents the numerical results at 3 dip levels of 7/10.5/14 dB. As expected, without SOA NL (low-power mode), the maximum gain achieves at full GE. At 14-dB dip level, 0.3 dB penalty is observed. It is due to the fact that the GE signal spectrum is not flat when adding the Tx noise, it creates colored noise that becomes larger at the higher GE dip. When SOA NL is

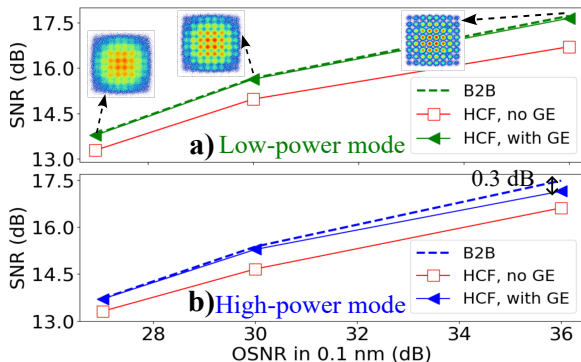


Fig. 4: Measured SNR versus OSNR (in 0.1 nm) without and with GE at (a) low-power and (b) high-power modes.

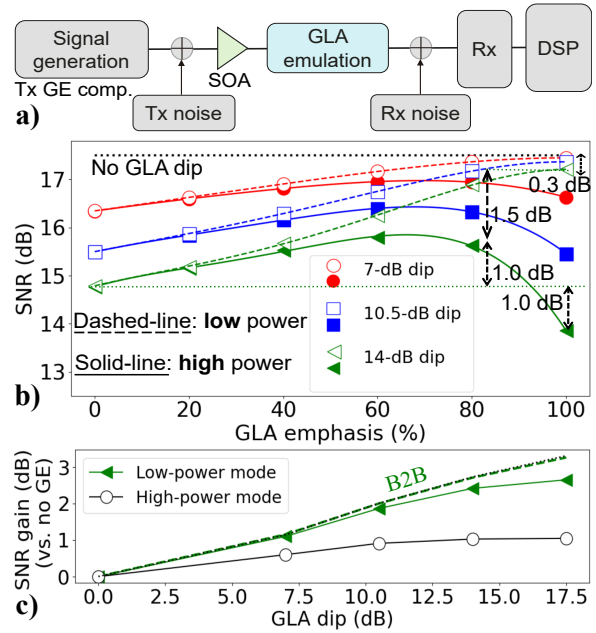


Fig. 5: (a) Simulation setup; (b) SNR versus GLA emphasis and (c) SNR gain for various dip values.

activated (high-power mode), 1 dB SNR gain is obtained at about 60% GE, which agrees well with the observation from experimental results (Fig. 3). At 14-dB dip level, up to 1.5 dB SNR penalty compared to maximum achievable SNR gain (of low-power mode) is observed even if using optimal GE of about 60%. It confirms again the interplay between SOA NL compensation and GE. The results also reveal that in the presence of SOA NL, using 100% GE can cause SNR penalty (i.e., 1 dB penalty with 14-dB dip). Finally, we presents the SNR gain of using GE (compared to no GE performance) at various dips in Fig. 5(c). Note that, the SNR gain here is taken at maximum achievable gain, i.e., at the GE ratio that optimizes the performance. At low-power mode, the SNR penalty is limited mainly by Tx colored noise inherent to combined GE and Tx noise (i.e., 0.3 dB and 0.6 dB penalties at 14 dB and 17.5 dB dips, respectively). In the presence of the interplay between SOA NL and GE (in high-power mode), the SNR penalty becomes bigger even with small dip of 7-dB. This penalty increases with the dip levels, i.e., up to 2.2 dB penalty at 17.5 dB dip. Our results therefore highlight the trade-off between the acceptable SNR penalty and the Tx output power, relying on the SOA power level and its NL.

Conclusions

We have numerically and experimentally compared GLA mitigation method, implemented at Tx and Rx sides, in 140 GBaud PCS-64QAM transmission systems. It is found that Tx side GE is suitable in the presence of dominant ASE noise. The partial GE has been demonstrated to provide a maximum gain when having the interplay between SOA NL and GE. The combination of both Tx and Rx GE remains for future studies.

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